

Limits:

$$\lim_{x \rightarrow 1} \frac{\ln(3x-2)}{x^2-1}$$

$$\Rightarrow \frac{\ln(3(1)-2)}{1^2-1} = \frac{0}{0} \quad \text{I.F.}$$

L'Hôpital Rule:

$$\frac{d}{dx}(\ln(3x-2)) = \frac{3}{3x-2}$$

$$\frac{d}{dx}(x^2-1) = 2x \quad \Rightarrow \quad \frac{\frac{3}{3x-2}}{2x}$$

$$\Rightarrow \frac{3}{3x-2(2x)} = \frac{3}{3x-4x} = \frac{3}{-x} = \frac{3}{-1}$$

$$\lim_{x \rightarrow 1} \frac{\ln(3x-2)}{x^2-1} = \frac{3}{-1}$$



$$\lim_{x \rightarrow -1} \frac{\sqrt{3+x^2} - 2}{\ln(x+2)}$$

$$\Rightarrow \frac{\sqrt{3+(-1)^2} - 2}{\ln(-1+2)} = \frac{\sqrt{3+1} - 2}{\ln(1)} = \frac{0}{0} \text{ I.F.}$$

L'Hôpital Rule:

$$\frac{d}{dx} (\sqrt{3+x^2} - 2) = \frac{2x}{2\sqrt{3+x^2}} \Rightarrow \frac{\frac{2x}{2\sqrt{3+x^2}}}{\frac{1}{x+2}}$$

$$\frac{d}{dx} (\ln(x+2)) = \frac{1}{x+2}$$

$$\Rightarrow \frac{2x(x+2)}{2\sqrt{3+x^2}} = \frac{2x^2 + 4x}{2\sqrt{3+x^2}} \quad \cancel{\frac{2x^2 + 4x}{2\sqrt{3+x^2}}}$$

~~Then~~ then: $\frac{2(-1)^2 + 4(-1)}{2\sqrt{3+(-1)^2}} = \frac{2 + (-4)}{2\sqrt{4}}$

$$\Rightarrow \frac{-2}{4} = -\frac{1}{2}$$

$$\lim_{x \rightarrow -1} \frac{\sqrt{3+x^2} - 2}{\ln(x+2)} = -\frac{1}{2}$$



$$\lim_{x \rightarrow e} \frac{1 - \ln x}{x - e}$$

$$\frac{1 - \ln e}{e - e} = \frac{1 - 1}{0} = \frac{0}{0} \quad \text{I.S.}$$

L'Hôpital Rule:

$$\frac{d}{dx} \left(\frac{1 - \ln x}{x - e} \right) = -\frac{1}{x}$$

$$\frac{d}{dx} (x - e) = 1 \quad \Rightarrow \quad -\frac{1}{x}$$

$$\Rightarrow \lim_{x \rightarrow e} \frac{1 - \ln x}{x - e} = -\frac{1}{e}$$



$$\lim_{x \rightarrow 3} \frac{\ln(x^2-8)}{x-3}$$

$$\Rightarrow \frac{\ln(3^2-8)}{3-3} = \frac{\ln(1)}{0} = \frac{0}{0} \quad \text{I.S.}$$

L'Hôpital Rule:

$$\frac{\frac{d}{dx}(\ln(x^2-8))}{\frac{d}{dx}(x-3)} = \frac{2x}{x^2-8} \Rightarrow \frac{2x}{x^2-8}$$

$$\frac{2(3)}{3^2-8} = \frac{6}{1} = 6$$

$$\lim_{x \rightarrow 3} \frac{\ln(x^2-8)}{x-3} = 6$$

